

Chronos — Comprehensive Technical Overview

A cinematic watch experience connected to WordPress and WooCommerce

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Live experience: Chronos

Source repository: Chronos on GitHub

Chronos combines an accepted cinematic frontend with verified CMS content, customer workflows and test payments. This reference explains the current experience, its engineering, the evidence collected and the additional work required for real commerce or larger audiences.

Current evidence and scope

Chronos is a connected WordPress and WooCommerce portfolio demonstration. The public catalogue, journal, pages and navigation read existing CMS records. Account login, customer order history, persistent contact intake and hosted Stripe test checkout are connected. Products and campaign visuals remain illustrative; no physical goods are sold and no live payment is accepted.

The connected release has 78 verified deployment files and 25 rendered HTML documents, with 20 indexable canonical URLs. The backend runtime matches local across 86 files. A hosted Stripe test payment was completed and reconciled with a paid WooCommerce order without reducing stock. Capacity of 10,000–1,000,000 readers and 99% availability remain objectives, not measured operating achievements.

What changed in this engineering pass

- Connected the accepted cinematic interface to eight published WooCommerce products, four WordPress posts and eight published pages, preserving existing records.
- Added editable information pages and navigation; verified administrator View links and actual Gutenberg editor save/reload.
- Restored account login and order history, persistent contact intake, authoritative test checkout and verified payment state.
- Fixed guest-cart loss, interrupted-session restoration, historical-receipt cart clearing and false contact success responses.
- Added safe WordPress HTML/media/link handling, genuine missing/draft HTTP 404 responses and a current CMS sitemap.
- Verified publication and editor changes in the browser without rebuilding; preserved marked test content as drafts afterward.
- Deployed locally verified code with before/after hashes, rollback protection and independent review; synchronized the project evidence and stable public reference.

1. Reading Guide and Evidence Policy

“Verified” means a named command or real flow passed and its evidence was retained. “Implemented” identifies source behavior; “target” identifies work still requiring operational proof. “Connected demo” means real CMS persistence and test payment integration, not a commercial launch. Legacy custom Watch, AI and analytics modules are distinguished from the storefront flows exercised here.

This edition supersedes the disconnected static-demo status. Visual polish alone did not establish WordPress integration; the current claims come from browser, API, administrator, editor, payment and deployment checks.

2. Executive Overview

Chronos combines a cinematic luxury-watch presentation with a functioning headless CMS. The frontend origin serves built HTML and assets; the browser retrieves current WordPress and WooCommerce data. Dynamic editorial/product URLs also consult the CMS for publication status before the origin serves them.

The design therefore has two operating boundaries: cacheable frontend delivery and authoritative CMS/database/API work. Login, contacts and test orders exercise persistent backend behavior. Neither the earlier static workload model nor the current functional checks establish commercial transaction capacity.

3. Product Purpose and Client Experience

Dark surfaces, warm accents, serif display typography, macro imagery and restrained motion express the watchmaking theme throughout the hero, collection, gallery, navigation, bag and journal. The accepted campaign design remains intact while actual CMS records populate the collection and editorial routes.

Visitors can browse, search, filter, sort, inspect products, manage a selection, sign in, inspect their orders, submit a contact message and enter clearly labelled test checkout. Keyboard controls, reduced motion, labels, static alternatives and responsive layouts support the experience. No shipping or real purchase is promised.

4. Verified Implementation Snapshot

- Content: eight published products, four posts and eight published pages; three verification records restored to drafts.
- Rendering: 25 HTML documents and 20 indexable canonical URLs; initial HTML refreshes at build time.
- Deployment: 78 of 78 frontend release hashes and 86 of 86 backend runtime-file hashes match local; archive restoration also passed.
- Connected browser suite: 23 checks; administrator: four checks; editor: two checks; publish-without-rebuild browser flow: five checks.
- Payments: hosted Stripe test payment confirmed paid in WooCommerce; stock unchanged; repeated receipt checks passed.
- Code: PHP 38 tests / 60 assertions; Gutenberg 19 tests; frontend types/build/release checks and installed source security scan passed.
- Warnings: frontend lint has zero errors and 12 existing warnings; PHPCS has zero errors and seven warnings. Both npm dependency audits report zero advisories.
- Distributed capacity, continuous uptime, complete manual accessibility assessment and commercial-launch compliance remain unverified.

5. Architecture Overview

Visitor → CDN/frontend origin → rendered HTML and versioned assets → React → GraphQL/REST → WordPress/WooCommerce → test payment provider. CMS publication checks protect dynamic routes; an up-to-date backend-generated sitemap exposes published URLs.

WordPress owns editorial records, navigation and contact persistence. WooCommerce owns product/stock/price/order truth. Validated local content owns cinematic assets, campaign copy and interface messages. Typed settings own public environment values; server-side settings own payment credentials and operational limits.

6. Route and Rendering Model

Existing public routes have build-time HTML, canonical/title/social metadata and readable content. Account and transaction routes are excluded from indexing. The origin validates dynamic CMS routes, returning actual 404 for missing or draft content and failing closed when the CMS cannot validate a route.

New publications and edits appear through browser API reads and the live sitemap without a frontend rebuild. Their initial HTML snapshot is not automatically regenerated: a newly published route initially receives the application shell, and existing snapshots may contain older content until the next build. This is an explicit rendering limitation, not server-side rendering or incremental regeneration.

Canonical index-file and trailing-slash aliases redirect with query strings preserved. Sensitive files and absent assets return 404. People and crawlers receive the same route response.

7. Frontend State and Query Ownership

UI state owns filters, gallery selection, dialogs and the local bag. React Query caches CMS responses and scopes private order/receipt queries to the authenticated identity. Catalogue and editorial adapters follow all cursors rather than assuming the first page is complete.

Guest selections survive sign-in; sign-out clears the active local token and selection. Storage failure has an in-memory fallback. A failed viewer request does not itself revoke a token; explicit authentication rejection does. Browser JWT storage remains sensitive to any successful script injection.

8. API Adapter Layer

GraphQL serves paginated products, posts, identity and order reads. REST supplies site configuration, pages, contact persistence and test checkout operations. Typed settings centralize endpoints, mode, timeout, page size and refresh interval; eight frontend environment variables have documented examples.

Responses are validated where they establish UI contracts or persistence acknowledgement. Only editorial page lookup treats 404 as missing content. Contact success requires a positive saved record ID; HTTP errors and malformed success payloads preserve the user's note. Connected content failure does not silently substitute the fictional local catalogue.

9. WordPress Editorial Ownership

WordPress owns published pages, posts, media, saved blocks and navigation. Administrator View links for published products/posts/pages resolve to the frontend. Four information pages and the primary menu were configured while preserving existing CMS content.

DOMPurify sanitizes editorial HTML before React inserts it. Media origins and internal links are normalized, and unsafe schemes/forms/scripts are excluded. Actual editor save/reload and publication checks established the path from maintained CMS content to rendered frontend content.

10. WooCommerce Domain Ownership

WooCommerce is the authority for product identifiers, raw prices, stock and order totals. The storefront displays the configured BDT currency. Checkout resolves products and availability on the server and rejects invalid quantities; browser display values never set the payable amount.

These are demonstration records, not verified manufactured inventory. Tax, shipping, fulfilment, consumer contracts and real merchant configuration need separate business requirements before any live-commerce launch.

11. Custom Bridge Plugin

Chronos Bridge 2.1 connects site/page/navigation APIs, contact intake, checkout/payment coordination and administrative functions. Its plugin header and runtime version agree. PHP tests, real authorization checks and deployed-file hashes support the current integration claims.

The recovery corrected server pricing, privileged order-field writes, payment completion acknowledgement and duplicate-attempt handling. A source module's presence is not proof that every legacy option, cron task or third-party service has been exercised.

12. Bridge Module Ownership

- Content/API modules expose site, published pages, sitemap, route status and GraphQL extensions.
- Commerce modules resolve server totals, coordinate test sessions and verify persisted order/payment state.
- Contact/privacy modules own persisted intake, administrator access and existing export/erasure integration.
- Operations modules retain cache, scheduled-maintenance and database lifecycle responsibilities; their complete schedule behavior was not load-tested.
- Custom Watch, analytics and AI-assisted administration remain retained capabilities outside the verified storefront path. No paid AI call was made.

13. Gutenberg Block System

Watch Showcase, Watch Collection Grid and Contact Form are registered in the actual WordPress editor. A marked draft saved a custom block and retained it after reload. All 19 unit tests, block lint and asset compilation passed; 17 changed source/build files were synchronized with exact hashes.

Block attributes remain persistent contracts. Future incompatible changes need migration planning. The earlier patched SockJS HTTP/WebSocket check remains maintenance evidence; this release adds real editor integration. A complete matrix of WordPress versions and every custom-watch rendering combination is not claimed.

14. Catalog and Editorial Data Flow

WooCommerce products and WordPress posts/pages → paginated GraphQL or validated REST → normalized media/links and sanitized content → collection, journal and detail routes. CMS navigation resolves through the same frontend origin.

A marked page, post and product were published after building. All three became readable, the new product/post joined their lists and the sitemap changed without rebuilding. An editor-updated page also showed its new text. They were restored to drafts, with missing-page behavior verified separately.

15. Cart and Session Flow

The browser bag records a requested selection and validates restored product/quantity data. It does not reserve stock. Login preserves a guest selection, while checkout revalidates every item against WooCommerce. Stale prices and availability cannot establish order authority.

Checkout retry state stores a stable request identifier and item fingerprint. A paid receipt clears only its matching purchase selection. Returning to an old receipt preserves a later bag; the repeated-receipt regression verifies that boundary.

16. Checkout and Payment Boundary

Checkout is connected in Stripe test mode only. Authenticated customers submit product IDs and quantities; the server calculates the order and creates a hosted test session. The browser only follows the canonical Stripe checkout origin. Live-session identifiers and unauthorized requests are rejected.

A hosted Stripe test card payment was completed, the provider reported a paid test session and WooCommerce persisted a paid processing order. Inventory stayed unchanged and demonstration order emails were suppressed. This is meaningful end-to-end test evidence, not settlement of real money or readiness to ship goods.

17. Payment State and Idempotency

A per-customer idempotency key, database lock and persisted order/session metadata coordinate retries. Return verification and signed webhook handling check ownership, session identity, amount, currency and provider payment state. The endpoint confirms persisted payment completion before reporting paid=true.

Repeated return verification converged on the same order. A different authenticated identity could not read the customer's payment result. Process interruption, provider/network failure and commercial reconciliation still require operational monitoring; functional replay tests do not establish every failure mode.

18. Contact and Customer Data Flow

Contact POST validates and rate-limits name, email, subject and message, then stores a record in the WordPress inbox. Actual administrator inspection confirmed persistence. Anonymous reads and unauthorized privileged operations were rejected. Missing/failed/malformed responses show failure and retain the unsent note.

Accounts, contact records and demonstration orders contain personal data. The published privacy page describes these flows and offers a contact route for access/correction/deletion requests. Contact intake does not send email; no outbound verification message or paid AI processing was performed. A real business must establish retention, lawful purposes, processor terms, transfers and operating rights-request procedures.

19. Search Metadata, Analytics, and Structured Data

The frontend retains semantic headings, canonical/title/description/social metadata, escaped structured data and discovery controls. Website/page/article metadata and connected Product metadata describe the CMS records without merchant offers; fictional campaign disclosures remain visible. The CMS sitemap reflects current published URLs.

Initial HTML snapshots refresh only on rebuild; client-rendered updates and new routes need JavaScript until new snapshots are generated. Search indexing and AI citations are not promised. Google's guidance does not require a special AI schema. Robots/noindex are discovery controls, not authorization. See [1]–[3].

No advertising or behavioral analytics integration was added. Public CSP and runtime checks guard unintended scripts; the no-transform policy remains in place.

20. Caching and Scheduled Maintenance

Hashed scripts/styles have immutable caching; media has a bounded lifetime and range support. HTML emphasizes freshness with no-transform. Error responses use no-store/no-transform. CMS/API and authenticated responses must not be treated as publicly cacheable personalized HTML.

The current connected path still reaches WordPress for browser data and dynamic route status. Cloudflare does not cache HTML by default. High-volume delivery needs measured anonymous cache behavior, safe invalidation and API/database capacity. Existing backend cache and scheduled modules do not establish those properties without runtime evidence. See [3].

21. Security and Privacy Boundaries

The frontend container serves only its built artifact, but the connected system now exposes CMS APIs and persistent account/contact/order flows. Authorization, input validation, server-owned prices, sanitized HTML, session ownership and payment verification therefore matter alongside static delivery controls.

CSP permits the configured backend for API/media traffic while restricting scripts to the same origin. Framing, workers, plugins and native form navigation remain restricted; inline styles support the component/motion system. The origin accepts GET/HEAD only and has sensitive-path, request-size and timeout limits. Backend APIs implement their own method/permission rules. See [4]–[5].

The non-root read-only frontend runtime drops capabilities and uses resource/log limits. Upstream HTTPS certificate verification remains enabled, with an explicit chain depth and bounded timeouts. These controls are not a full host penetration test, perfect bot protection or complete CDN-origin isolation.

22. Failure Behavior and Recovery

Media has static/poster fallbacks and reduced-motion handling. Denied storage avoids import-time crashes. Missing/draft routes return 404; CMS lookup failure fails closed. Contact failures retain text, interrupted viewer requests preserve the token until explicit rejection, and old payment receipts preserve later selections.

Immutable frontend releases and backend before-snapshots support rollback. Promotion verifies prior state, candidate behavior and final hashes; archive restoration was checked. One render attempt timed out on a backend page request and the retry passed, so backend latency remains an operational consideration rather than being hidden as a clean first run.

23. Test and Verification Evidence

- PHP: 38 tests / 60 assertions passed; PHPCS zero errors and seven warnings.
- Gutenberg: 19 tests, lint/build and actual editor registration/save/reload passed; both npm audits report zero advisories.
- Frontend: types, client build, 25-page render, release verification and source security scan passed; lint zero errors and 12 existing warnings.
- Connected built-preview: 23 browser checks; actual administrator: four; editor: two; newly published content: five.
- Persistence/payment: real hosted test payment, unchanged stock, repeated receipt three checks, account/inbox persistence two checks and authorization/input eight checks passed.
- Contact failures: missing endpoint, server error and malformed success all retained the note and displayed failure.
- Delivery: isolated actual Nginx route/header/alias/404/range/sitemap checks passed; deployed frontend 78/78 and backend runtime 86/86 hashes match.
- Independent source review passed after receipt, authentication and contact error handling corrections. Public-domain verification passed 16 content/login/responsive checks, three focused login/logout checks and 12 HTTPS checks; six accessibility scans recorded zero violations and the browser recorded zero runtime errors.

24. Test Evidence Interpretation

The evidence covers the connected demonstration and named flows. Earlier cinematic/static tests remain historical design evidence, not a substitute for current CMS/API checks. Test payment success does not certify real fulfilment, live webhook operations or all provider failure modes. Automated accessibility scans do not replace manual assistive-technology assessment.

No real secrets were found in the scanned changed-source scope. Full-history scanning returned two vendored JWT documentation keys; exact upstream versions contain the same public examples. The raw history scan is not described as zero findings. Dependency audit counts are dated observations, and lint/PHPCS warnings remain visible.

25. Workflow and Deployment Discipline

Local code is the source of truth. Live files are inventoried and compared before changes; local-ahead changes are validated, deployed and rehashed. Runtime files match; local CI tests are not deployed as application code. Immutable frontend releases retain the previous release and automatic failure recovery.

GitHub workflows remain manually guarded and were not triggered as deployment evidence. Current frontend delivery uses the existing shared server; the WordPress backend uses its existing host. Historical workflow targets are not interchangeable with the current release process. Final repository changes are committed locally, with pushing left to the maintainer.

26. Operational Observability

The availability helper checks expected HTTPS response content and records timestamps; runtime logs are bounded. A static health endpoint alone does not establish CMS health, so the connected operating plan also needs catalogue/API, login and test-payment observations with appropriate non-destructive probes.

No independent always-on scheduler or continuous 30-day record was established. Monitor placement, missing-data treatment and escalation ownership remain necessary. Private checkpoints record before/after release identity, pending commands and recovery actions for power-loss handoff.

27. Verified Limitations and Risk Register

- Current frontend and CMS each retain host/database failure boundaries; no independently verified automatic failover is claimed.
- Large-reader capacity and 99% availability are targets; connected API and route-status traffic require new workload measurements.
- Initial HTML snapshots refresh on rebuild; new publications rely on browser rendering until then.
- Browser JWT storage is sensitive to script compromise; a future commercial session design needs separate review.
- Real merchant identity, processor contracts, retention, taxes, shipping, refunds and market/product facts are not supplied by this portfolio.
- Paid AI, outbound email, comprehensive legacy cron/analytics behavior, manual full accessibility review and host penetration testing were not exercised.
- Existing lint/PHPCS warnings remain documented; zero dependency advisories is not universal security certification.

28. Current Status Summary

The cinematic frontend is connected to WordPress content, WooCommerce records, account/order history, persistent contact intake and Stripe test checkout. Actual administrator/editor, browser/API, payment and release-parity evidence supports those statements.

Commercial operation, large-scale capacity and continuous availability remain separate objectives. The project demonstrates the tested full-stack path while identifying the boundaries that still require business decisions and operational proof.

29. Developer Handoff

Start with README, verification, scalability and security documentation. Use lockfiles and the documented Node/PHP versions; run types, lint, tests, audits, build and release verification. Configure connected endpoints centrally; maintain CMS records in WordPress and cinematic/UI data in validated files. Never place privileged values in public build variables.

Trace a product through server price/stock resolution, authenticated order creation, idempotent test checkout and verified payment state. Trace an editor save/publication through safe frontend rendering and actual HTTP status. Trace a contact acknowledgement to its admin record. Keep credentials and operational recovery private; update the dossier before deriving public documentation.

30. Cinematic Frontend and Motion Engineering

The redesign uses macro watch imagery, campaign films, editorial rhythm and intentional product close-ups. Scroll-linked film seeking supports forward and reverse movement. Responsive variants, self-hosted fonts, optimized images, video posters and motion preferences support a usable experience when autoplay, video loading or animation is unavailable.

Product cards, navigation, buttons and the bag drawer follow the same visual language. Gallery failure recovery, focus return, menu containment and live reduced-motion updates were explicitly exercised. AI-generated imagery and films are disclosed as fictional illustration; asset provenance and font/software licences remain part of the handoff. Technical polish is not presented as evidence that the fictional watch specifications were manufactured or tested.

31. Dependency and Configuration Maintenance

The frontend moved to the updated Vite 7 and React Router 7 toolchain. Gutenberg tooling uses WordPress scripts 35 with refreshed packages. Patched transitive resolutions address advisories without suppressing audit failures or taking npm's suggested obsolete toolchain downgrade. Both npm trees report zero advisories at the recorded check.

Configuration ownership is explicit: typed settings for public environment values, validated content files for maintained product/copy/search data, and deployment configuration for runtime policy. JSON-LD serialization escapes unsafe delimiters. Private keys, real secrets, environment files, recovery records and personal test evidence are excluded from the public release. Eight public frontend variables are represented in typed settings and the safe environment example. Server payment settings and maintained form/UI content have explicit owners; no real secret was found in the changed-source audit.

32. US and EU Privacy: Applied Demo Measures

The current privacy notice describes sign-in tokens and local bag storage, WordPress account/contact/order records, test payment processing and provider request metadata. Sign-out clears the active local token and selection. Visitors are asked not to reuse important passwords or submit sensitive information. Data-rights requests use the contact route. A retention/deletion operating schedule and processor arrangements must be established before real business use.

CNIL guidance distinguishes strictly necessary user-requested functionality from consent-requiring trackers. This is not a blanket exemption for all local storage. The demo adds no advertising or behavioral analytics. Any future tracking feature requires reassessment before loading. See [7]–[8].

GDPR applicability depends on establishment, offering goods/services to people in the EU or relevant monitoring, rather than nationality alone. Where applicable, controller identity, lawful purposes/bases, recipients, retention, rights, transfers, processor arrangements and security/incident responsibilities must be established. The demo does not invent a controller address, EU representative or DPO. See [9]–[10].

US state privacy rules depend on business and processing facts. California's published adjusted revenue threshold is \$26,625,000, alongside other applicability tests; visitor count alone does not decide coverage. Other state laws and child-directed collection require separate review. The current demo has no sale/sharing-for-advertising feature; that is not a future consent-management platform. See [11].

33. Consumer, Product, Accessibility and AI Law

US consumer protection: fictional products and AI visuals are disclosed; fabricated reviews, testimonials and unsupported performance claims are excluded. Real merchandise sales require substantiated shipping promises and operational delay/cancellation/refund handling under the FTC's rules. Email marketing would require CAN-SPAM compliance and applicable regional consent analysis. No shipping or unsubscribe service is simulated. See [12]–[14], [18].

EU consumer protection: real distance sales require trader identity, product and total-price information, delivery/complaint arrangements and appropriate withdrawal information. The general withdrawal period is 14 days, subject to exceptions; goods have statutory guarantee rights distinct from optional warranties. These are documented launch obligations, not demo checkout promises. See [15]–[16].

Product safety: the GPSR introduces responsibilities and online-offer information requirements. Real watches need product classification, manufacturer/responsible-person information and applicable warnings; materials, batteries, connected features, waste rules and national requirements need facts and evidence. Tax/VAT, import duties, sales-tax nexus, trademarks and shipping markets also require business review. No compliance badge is used as a substitute. See [17].

Accessibility: US ADA and EU accessibility scope are considered without claiming that a government website rule automatically governs every private demo. Keyboard controls, semantic content, labels, alternatives, reduced motion and contrast improvements are implemented. Automated scans found no violations in the tested pages; manual screen-reader assessment and applicable national-law review remain necessary. See [19]–[20].

Synthetic media: visible fictional/AI disclosures support transparency. EU AI Act duties vary by provider/deployer role and content; no claim of cryptographic provenance or full provider compliance is made. US copyright protection depends on human authorship; generated assets are not described as universally exclusive or fully cleared. Preserve model terms, source licences and provenance before commercial reuse. See [21]–[22].

34. Architecture for 10k–1M Simultaneous Readers

The frontend assets and build snapshots can be replicated, but the active browser catalogue, identity/contact/payment operations and dynamic route checks depend on WordPress and its database. The previous static model is therefore only an asset-delivery scenario. It is not a measured capacity model for this connected release.

The maintained illustrative workload assumes a 300-second session, five HTML requests, 25 asset requests and 12 MB transfer. Its arithmetic gives about 1,000 edge requests/second at 10,000 sessions and 100,000 requests/second plus 320 Gbps at 1,000,000 sessions. These are assumptions, not benchmarks or purchased capacity. They omit newly active API/route-status traffic.

Even a hypothetical 99% cache-hit ratio leaves substantial origin traffic; the present 100 requests/second frontend guard does not provide million-reader capacity. A real plan needs anonymous API caching/invalidation, bounded database connections, route-status caching with publication correctness, media bandwidth, independent origins and CMS/database resilience. Personalized and transactional data must stay out of shared caches.

Validation stages remain bounded local tests, isolated representative origin/API tests, distributed 10k, 100k soak/cold-cache/failover and a separately funded 1M test. Define geography, think time, API mix, cache state, latency/error thresholds and recovery. No shared-server load test or paid load service was launched.

35. 99% Uptime Objective and Evidence Plan

The objective is 99% end-to-end availability over a rolling 30-day window. That corresponds to a 432-minute error budget, or 7 hours 12 minutes. Success means expected HTTPS content within the configured timeout, not merely an open network port.

The probe supports a configured 60-second interval when run by an independent scheduler. Reporting must include observation coverage and treat missing intervals as unknown. A handful of successful manual checks is not a measured uptime percentage. No continuous 30-day record or contractual SLA is claimed.

A complete resilience demonstration requires a second failure domain, matching release hashes, controlled traffic failover, measured recovery time and safe primary restoration. Health checks, process restart and rollback are implemented foundations, but cannot remove a single-host outage by themselves. Any additional paid hosting, routing, monitoring or traffic commitment requires separate approval.

Appendix A. Representative End-to-End Traces

Browsing: WooCommerce/WordPress records → paginated API → validated/sanitized view → collection, journal or page → local selection.

Editorial: actual editor save → publication → current route validation and sitemap → browser fetch; rebuild later refreshes initial HTML.

Test commerce: authenticated product IDs/quantities → server totals/stock → idempotent order/session → hosted test payment → owner/amount/currency verification → persisted paid order.

Contact: validated POST → saved inbox ID → success acknowledgement → administrator-only record inspection; errors preserve the draft.

Release recovery: snapshot/compare → local edit/gates/review → isolated candidate → guarded promotion → public checks and hashes → previous release retained.

Appendix B. Data and Authority Matrix

- Campaign visuals, fictional concepts and UI messages: validated maintained local content.
- Editorial pages/posts, menu and saved block attributes: WordPress.
- Products, prices, stock and demonstration orders: WooCommerce.
- Payment status: test provider verified against the owning WooCommerce order.
- Contact records: backend persistence with administrator access.
- Filters, dialogs and temporary selection: browser state; never financial authority.
- Public endpoints and limits: typed configuration; privileged values remain private/server-side.
- Capacity and uptime claims: measured evidence; design objectives remain labelled.

Appendix C. Developer Review Checklist

- Run frontend and block audits, types/lint/build/release checks, PHP tests/standards and source scanning.
- Exercise real CMS products/posts/pages, editor save/publication, login/logout, order history, contact persistence and test checkout.
- Include negative authorization, missing/malformed persistence responses, repeated receipts and interrupted navigation.
- Verify draft 404, published routes, sitemap, CSP/media permissions and responsive/accessibility behavior.
- Preserve local-first drift checks, candidate validation, rollback and exact final runtime hashes.
- Reassess session design, retention, provider failure recovery and business obligations before real commerce.
- Keep dossier, public reference, canonical PDF and private recovery checkpoints synchronized.

Appendix D. Client Review and Acceptance Checklist

- Inspect the live cinematic presentation, collection, journal, information pages and mobile layout.
- Use the separately supplied private administrator access to inspect products, posts, pages, contact records and test orders.
- Confirm CMS changes appear in the frontend; understand the separate rebuild requirement for initial HTML snapshots.
- Use only test credentials/card details in checkout; no real goods or live charges are offered.
- Review actual verification results and remaining commercial/operational requirements.
- Treat large concurrency and 99% uptime as objectives until representative evidence exists.

Appendix E. Primary Sources and Further Reading

The following primary sources support the legal and technical analysis. Applicability depends on the real business and deployment facts; this report does not certify universal US/EU compliance.

- [1] Understand JavaScript SEO basics
- [2] AI features and your website
- [3] Default cache behaviour
- [4] Content Security Policy Cheat Sheet
- [5] HTTP core module; request limit module
- [6] Unexpected external redirect advisory; SSR constructor injection advisory
- [7] Cookies et traceurs: que dit la loi?
- [8] Cloudflare cookies
- [9] Guidelines 3/2018 on territorial scope
- [10] Regulation(EU)2016/679, GDPR
- [11] Adjusted monetary thresholds; FAQs
- [12] Advertising and marketing
- [13] Final rule banning fake reviews and testimonials
- [14] Business guide to the Mail, Internet, or Telephone Order Merchandise Rule
- [15] E-commerce, distance and off-premises selling
- [16] Consumer guarantees
- [17] General Product Safety Regulation summary; GPSR questions and answers
- [18] CAN-SPAM compliance guide
- [19] Guidance on web accessibility and the ADA
- [20] Accessibility of products and services
- [21] Transparency obligations under Article 50 of the AI Act; Regulation(EU)2024/1689
- [22] Copyright and artificial intelligence

Appendix F. Repository References

Verified release results

Scalability and availability architecture

Security operations

Search, security and legal research

The public deployment is verified. Newly prepared repository documentation becomes available at these branch links after the maintainer pushes the local commit.

Developed by Zahidul Islam • Cinematic frontend craft, explicit system boundaries and verifiable engineering.